

ANISH SHIRODKAR

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SUMMARY

MS Computer Science candidate at Rutgers University specializing in ML and AI engineering. Research spans clinical EHR transformers (BEHRT), information retrieval (bi-encoder + reranker), RAG pipelines, and full-stack LLM applications — shipped production AI systems end-to-end. Seeking an AI/ML Engineer Intern role for Summer 2026.

EDUCATION

Rutgers University-New Brunswick

September 2025 – Present

MS Computer Science, AI & ML Track | GPA: 3.75/4

Thakur College of Engineering & Technology, University of Mumbai

December 2021 – May 2025

B.Tech IoT | CGPA: 9.50/10 | Ranked 2nd in the batch

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C, SQL

AI & Machine Learning: PyTorch, TensorFlow, Scikit-learn, LSTM, Attention Mechanisms, Transformers, Fine-tuning, PPO, DQN

NLP & Information Retrieval: LangChain, RAG, Sentence Transformers, HuggingFace, ChromaDB, Qdrant, BM25, Cross-encoder Reranking, NDCG, MRR, Vector Databases

Generative AI: Groq/LLaMA 3.1, LLaMA 3.3 70B, Gemini API, Prompt Engineering

Computer Vision: YOLOv8, ByteTrack, OpenCV, Ultralytics, Object Detection, Multi-Object Tracking

Web & Backend: React, Next.js, FastAPI, REST APIs, Supabase (PostgreSQL + RLS), Firebase

MLOps & Infrastructure: Docker, Model Deployment, Inference Optimization, Hugging Face Spaces, Vercel, AWS, Weights & Biases, Git, CI/CD

EXPERIENCE

MIARC Lab, Rutgers University | *AI/ML Research Intern*

New Brunswick, NJ | June 2026 – Present

- Built BEHRT-style clinical transformer over 540K+ patient RWJBH EHR cohort on the Amarel HPC cluster (A100), integrating 6 clinical domains — CCSR diagnoses, medications, labs, vitals, procedures, and orders — under Dr. Branimir Ljubic.
- Engineered patient-level preprocessing pipeline across 31.6M comorbidity rows and 24.1M medication rows; normalized 8 antihypertensive drug classes via generic + brand-name matching and mapped ICD codes to 513 CCSR categories.
- Trained 4.66M-parameter transformer (vocab 12,814) with masked language modeling; loss reduced 8.96 → 6.29 over 15 epochs, establishing baseline for downstream hypertension risk prediction.

Indian Meteorological Department | *ML Project Member*

Gujarat, India | April 2023 – June 2025

- Led end-to-end ML model development for a live government weather forecasting system; built IoT-to-cloud data pipeline delivering real-time predictions at production scale.
- Engineered attention-LSTM hybrid that improved multi-regime forecast accuracy across seasonal cycles; validated and registered as government copyright (IP India, No. LD-20250175526, 2025).

PROJECTS

Two-Stage Semantic Search | *sentence-transformers, Qdrant, cross-encoder, FastAPI, MS MARCO*

2026

- Engineered two-stage retrieval pipeline (bi-encoder MiniLM-L6 + cross-encoder reranker) over 100K MS MARCO passages; achieved NDCG@10 of 0.692 — 143% above BM25 and 12% above bi-encoder alone.
- Designed NDCG@10, MRR@10, Recall@100 evaluation harness benchmarking 3 retrieval systems end-to-end; deployed production API via FastAPI + Qdrant with Apple Silicon MPS acceleration, containerized with Docker.

Vouch: Cross-Party AI Agent Trust Layer | *Auth0 Token Vault, Groq/LLaMA 3.3 70B, React, Express*

2026

- Architected production-grade credential delegation system using Auth0 Token Vault; enables multi-agent AI workflows with scoped, auditable access tokens — eliminating raw credential exposure across trust boundaries.
- Delivered Groq/LLaMA 3.3 70B reasoning engine, Vouch SDK CLI, and React/Vite dashboard with role-based access control; deployed full-stack to Render with Express API backend serving live agent requests.

SkillGap: AI Career Assistant | *Next.js 15, React 19, Supabase, Gemini 2.5 Flash, pdf.js, shadcn/ui*

2026

- Shipped two-stage NLP pipeline using Gemini 2.5 Flash to extract and rank skill entities from PDF resumes and JDs; validated gap report accuracy and ranking quality across 15+ real-world resumes.
- Built stateful multi-turn AI interview simulator leveraging prompt grounding; generates role-specific technical questions anchored to candidate resume with structured performance scoring.

CERTIFICATIONS AND PUBLICATIONS

Certifications: AI Agents Fundamentals (Hugging Face), MCP: Advanced Topics (Anthropic), Introduction to Agent Skills (Anthropic) — March 2026

Publication: Shirodkar, A. V. *Smart Weather Forecasting: IoT-Integrated Decision Support System for Real-Time Analysis*. IP India Copyright, No. LD-20250175526, 2025.